

EP3120 - Statistical Mechanics

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Logistics

- Contact:
 - ▶ Google classroom class code - will be shared by email
 - ▶ kdmakwana@phy.iith.ac.in
 - ▶ PH-212
- Statistical Mechanics by Kerson Huang, 2nd edition
- Fundamentals of Statistical & Thermal Physics by F. Reif
- Statistical Mechanics by R. K. Pathria
- Lecture notes by Prof. Matthew Schwartz, Harvard University
<https://scholar.harvard.edu/schwartz/teaching>
- Lecture slides will also be provided

Evaluation

- Assignments - ungraded but useful as practice for exams
- 20% - first exam
- 20% - second exam
- 20% - Class project
- 40% - final exam

Course contents

Thermodynamics, phase space of classical ideal gas, Boltzmann's H-theorem, Maxwell-Boltzmann distribution, microcanonical ensemble, Gibbs paradox, Sackur-Tetrode entropy of ideal gas, canonical and grand-canonical ensembles, thermodynamics from partition function, statistical mechanics of simple harmonic oscillators (classical and quantum), equipartition theorem, para-magnetism, solid-gas equilibrium, quantum statistics (fermions and bosons), Fermi gas, Chandrasekhar mass limit, Bose gas, Planck's radiation law, Debye model of heat capacity, Bose-Einstein condensation, Ising model in 1D and 2D.